

Lecture 20: Change of Variables & the Jacobian

Background & Motivation

We've used polar, cylindrical, and spherical coordinates with extra factors of r or $\rho^2 \sin \phi$ in the integrand. Today we explain where those factors come from: the Jacobian determinant, which measures how a coordinate transformation stretches or shrinks area and volume. This also lets us design custom substitutions for specific regions.

1. Warm-Up: Where Does $r \, dr \, d\theta$ Come From?

In polar coordinates, $dA = r \, dr \, d\theta$. We have used this factor of r many times – today we explain *why* it appears.

Evaluate $\int_0^{2\pi} \int_0^3 r \cdot r \, dr \, d\theta$ to find the area of a disk of radius 3.

1. Inner (r): _____ (using $f = 1$, so integrand is just r from dA)
2. Outer (θ): _____
3. Check: $\pi(3)^2 =$ _____ ✓

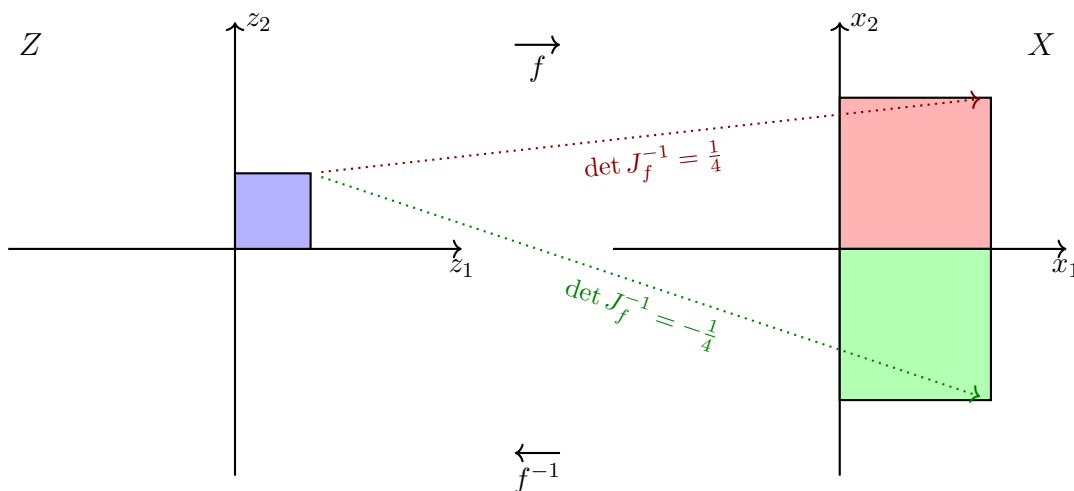
2. The Jacobian Determinant

Definition 1: Jacobian (2D)

For a transformation $T: (u, v) \mapsto (x, y)$, the **Jacobian** is:

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

- $|J|$ measures how much the transformation stretches or shrinks area at each point
- $dA_{xy} = \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du \, dv$



How the Jacobian determinant measures area scaling under a transformation. Different parts of the source region Z get scaled by different factors when mapped into X — the absolute value of $|J|$ gives the local scaling factor, and a negative determinant indicates a reflection.

Definition 2: Jacobian (3D)

For $(u, v, w) \mapsto (x, y, z)$:

$$\frac{\partial(x, y, z)}{\partial(u, v, w)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix}$$

– Same idea: $dV_{xyz} = |J| du dv dw$

3. Change of Variables Formula

Theorem 1: Change of Variables (Double Integral)

$$\iint_R f(x, y) dA = \iint_S f(x(u, v), y(u, v)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du dv$$

where S is the region in the uv -plane corresponding to R .

Triple integral version:

$$\iiint_R f dV = \iiint_S f(x(u, v, w), \dots) \left| \frac{\partial(x, y, z)}{\partial(u, v, w)} \right| du dv dw$$

4. Examples

Example 1: Polar Jacobian Verification

Verify that the Jacobian of $(r, \theta) \mapsto (r \cos \theta, r \sin \theta)$ is r .

Example 2: Custom Substitution for a Parallelogram

Evaluate $\iint_R (x - y) dA$ where R is the parallelogram with vertices $(0, 0)$, $(2, 1)$, $(3, 3)$, $(1, 2)$.

Example 3: Spherical Jacobian Verification

Show that the Jacobian for spherical coordinates is $\rho^2 \sin \phi$.

Example 4: Ellipse Area via Substitution

Use a change of variables to show that the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is πab .

Practice Problems

Problem 1. Compute the Jacobian $|J|$ of the transformation $x = u + v$, $y = u - v$.

Problem 2. Verify the cylindrical coordinate Jacobian: $x = r \cos \theta$, $y = r \sin \theta$, $z = z$. Show $|J| = r$.

Problem 3. Use $u = x + y$, $v = x - y$ to evaluate $\iint_R (x^2 - y^2) dA$ where R is the square with vertices $(1, 0)$, $(0, 1)$, $(-1, 0)$, $(0, -1)$.

Hint: $x^2 - y^2 = (x + y)(x - y) = uv$ and the region in uv -space is the square $|u| \leq 1$, $|v| \leq 1$.

Problem 4. Use $x = 2u$, $y = 3v$ to evaluate $\iint_D xy \, dA$ where D is the region inside the ellipse $\frac{x^2}{4} + \frac{y^2}{9} = 1$, first quadrant only.

Problem 5. Let $T: (u, v) \mapsto (u^2 - v^2, 2uv)$. Compute the Jacobian and describe geometrically what this transformation does. *Hint:* think complex numbers, $z = u + iv \mapsto z^2$.